An integrated scRNA-seq lung atlas built from agent-curated datasets (TODO: Be consistent with "cell" vs "barcode")

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Abstract

The abundance of nucleotide sequence data has grown into the order of trillions of bases. As public archives grow in scale, so too does their fragmentation and complexity, meaning that disease-focused meta-analyses remain difficult to accomplish. We identified the need for producing atlas-scale datasets that both incorporate work spanning studies and are uniformly processed and tissue-targeted. scBaseCount hosts uniformly processed single-cell RNA sequencing files, but those files are not merged, batch corrected, or clustered. To address these shortcomings, we pulled lung datasets from scBaseCount, applied quality control, and combined them into one atlas. Concatenation exposed a study-level batch effect, but scBaseCount contains no batch variable coarser than SRA experiment accession. We therefore retrieved National Center for Biotechnology Information's BioProject accessions from the European Nucleotide Archive and used them as the Harmony batch key. Finally, we clustered the neighborhood graph across a range of resolutions and selected the resolution that maximized the summed Jaccard index of the optimal one-to-one matching between clusters and published cell-type labels. The atlas contains 9,307,963 cells from 1,410 Sequence Read Archive experiment accessions spanning 172 BioProject accessions. Follow-up checks showed that cells from different studies mixed more after batch

correction, while cell-type structure was largely retained. The scale and integration of the clustered dataset presents a possibility of making novel cross-study findings, even across rare cell types, through downstream analysis.

Introduction

Public single-cell RNA sequencing (scRNA-seq) archives now contain enough cells for tissue-scale reuse, but experiments are still deposited independently, processed with different pipelines, and described with free-text metadata (Luecken et al., 2022; Mueller et al., 2026). Cross-study analysis therefore rarely starts from a shared gene axis or a study-level batch structure. Building an atlas from public data remains a separate construction problem, whereby a tissue must be selected, matrices must be brought onto a common reference, and the joint object must be integrated without erasing cell-type biology.

Community contributed collections such as CZ CELLxGENE Discover enforce cell-level and metadata standardization on submission, allowing data exploration and meta-analysis (CZI Cell Science Program et al., 2025; Rozenblatt-Rosen et al., 2017). Efforts like the Human Lung Cell Atlas (TODO: distinguish from Human Cell Atlas) integrate selected respiratory studies into a consensus cell-type map (Sikkema et al., 2023). These resources straddle the inevitable tradeoff between being exhaustive and curated. In the case of the mentioned efforts they do not ingest the Sequence Read Archive (SRA) as a whole (Leinonen et al., 2011). On the other hand, scBaseCount opts for extensiveness by using an agentic AI metadata curation approach to collect and reprocess public 10X scRNA-seq experiments at large scale, producing a library of `h5ad` files keyed by SRA experiment accession (Youngblut et al., 2025). Those files do not contain usable clusters. They are also neither tissue-restricted nor treated with any batch correction.

Here we construct an integrated lung-tissue atlas from the `2026-01-12` scBaseCount human GeneFull release. We focus on lung tissue because of the frequency and severity of lung diseases. Lung cancer was estimated to be the most frequently diagnosed cancer at 2.6 million new cases, and the leading cause of cancer death with 1.9 million deaths in 2024 (Sung et al., 2026). Non-cancer diseases such as idiopathic pulmonary fibrosis are also a cause for concern and are known to be associated with high mortality rates (Chakraborty et al., 2022). scRNA-seq has promise and meaningful progess when it comes to researching lung diseases such as these (Chakraborty et al., 2022; Navin et al., 2011). To focus on lung tissue samples, we select experiments from scBaseCount whose tissue labels match lung. We apply per-dataset quality control, align datasets by gene and concatenate them over cell them into a single object. We determine that scBaseCount does not contain any cross-dataset variable that could serve as a key to correct for the batch effect. Therefore, we retrieve the National Center for Biotechnology Information's (NCBI) BioProject accessions for each dataset and use them as the Harmony batch variable (Barrett et al., 2012; Korsunsky et al., 2019). (TODO: mention that context lookup produces a json object for each SRX–here there is further information for stuff like disease state, etc) We also develop a method to optimize a clustering algorithm parameter, namely the Leiden algorithm's resolution parameter, by matching clusters to CZ CELLxGENE `cell_type` labels (Traag et al., 2019). Finally, we evaluate batch correction with a suite of metrics proposed by Luecken et al. (2022) on a study-stratified subset. The result is an analysis-ready lung atlas derived from an agent-curated public resource.

Materials and methods

Building an integrated lung atlas using scBaseCount data required developing robust single-cell workflows in Python using Scanpy (Wolf et al., 2018). We enforced observability through extensive logging configured with the Python logger.

Data acquisition

scBaseCount data, which includes expression matrices, are hosted on Google Cloud Platform in Google Cloud Storage (GCS) buckets (Youngblut et al., 2025). We used human, GeneFull matrices from the `2026-01-12` release of scBaseCount. Before downloading any matrices, we identified a subset of the available matrices from corresponding metadata table shipped with scBaseCount. We discarded accessions that were not SRA or ENA records (identifiers beginning with NRX). We then flagged lung tissue by applying a regular expression to the free-text tissue field, matching terms such as "lung," "pulmonary," "alveolus," and "bronchus." For each remaining accession we retrieved a BioProject accession from the European Nucleotide Archive (ENA) portal API (Amid et al., 2020). We excluded seven BioProjects known to mix lung with many other organs (PRJEB51634, PRJNA1005589, PRJNA1179423, PRJNA1188170, PRJNA1215450, PRJNA657844, and PRJNA902813), dropped accessions that lacked a BioProject accession, and dropped studies whose combined cell count was 1,000 or fewer. We sequentially fetched datasets that remained after these filters from GCS and computed their MD5 hashes on disk. Due to the extensive cumulative size of the datasets, we uploaded the datasets themselves, along with the newly computed hashes, to an internally owned and maintained Cloudflare R2 bucket, serving as a production mirror of scBaseCount, allowing us to remove local files from disk.

Atlas construction and quality control

To integrate the aquired individual datasets as a single atlas-scale `h5ad` file, we built a pipeline to sequentially download scBaseCount datasets from the R2 mirror, filter them by performing quality control operations, determine if what remained of the filtered datasets was sufficient for inclusion in the atlas, align the datasets to a fixed gene axis, and concatenate the datasets to one in-memory atlas. We wrote the concatenated atlas to disk on pipeline completion.

To ensure datasests matched scBaseCount, we computed dataset MD5 hashes following download from R2 and compared them the MD5 hashes stored as metadata alongside the corresponding R2 objects, the latter having been computed upon upload. We enforced identity between the upload and download hashes.

Next, we filtered cells based on tunable parameters. We did not drop genes for rarity. We flagged genes as likely mitochondrial, ribosomal, and hemoglobin genes by applying regular expressions to the gene names, and computed per-cell quality metrics. We first dropped cells with a total gene count below 200 genes. We dropped cells whose mitochondrial and ribosomal gene counts were at least 20% and 50%, respectively, of total gene counts. We did not filter cells on hemoglobin gene counts. Following these filters, we applied a cutoff on number of remaining cells and percentage of cells remaining relative to the initial number of cells in the dataset. We did not include datasets in the atlas if they had fewer than 100 cells remaining or if fewer than 50% of cells remained in the dataset following filtering. We also excluded datasets in which every cell lacked a usable CZ CELLxGENE `cell_type` annotation (CZI Cell Science Program et al., 2025). We configured every mentioned threshold as a tunable parameter. To ensure that the values we settled on were well calibrated, we recorded the number of cells dropped by each filter for every file that reached QC, and we stored those counts both for all QC-processed files and for the subset of files that were concatenated, allowing us to rerun the filtering pipeline with different filtering parameters. We did not apply doublet detection. We chose filtering parameters based on recommendataions proposed by Luecken and Theis (2019) and enriched them with the ribosomal filter and hemoglobin flagging. Before we concatenated the filtered datasets along the cell axis to the atlas `h5ad` file, we reindexed each dataset onto the scBaseCount `geneInfo.tab` gene list, discarding genes absent from that list and filling missing reference genes with zeros, to ensure concatenated matrices used a shared gene axis. Since Youngblut et al. (2025) already indexed scBaseCount matrices against a shared STAR reference, this reindex did not change the genome reference.

Embedding and batch correction

We developed a method to embed and cluster the atlas using the CZ CELLxGENE `cell_type` key present in the datasets. First, we normalized gene counts per cell so that each cell had the same total gene count. Then, we logarithmized the data matrix by the natural logarithm, selected the top 2,000 highly variable genes (HVG) stratified by BioSample—a number in the range of recommended HVGs according to Luecken and Theis (2019)—and z-scaled the expression of each gene. We then applied principal component analysis (PCA), computing the first 50 PCs. We then recorded the minimum number of PCs in the range of 15–50 explained at least 50% of the data's cumulative variance. We applied the processing steps through PCA using Scanpy (Wolf et al., 2018).

Following PCA, we applied Harmony to the PCs to correct for one batch effect (Korsunsky et al., 2019). The scBaseCount release from which we collected data did not contain any keys coarser than SRA experiment accession, which was effectively a one-to-one file key. We identified the NCBI BioProject accession as a key to use for applying batch correction to the atlas. BioProject accessions group SRA experiment accessions by project type, project scope, technical methodology, or other groupings (Barrett et al., 2012). To collect the BioProject accession for a given SRA experiment accession, along with other relevant metadata, we queried the European Nucleotide Archive's (ENA) portal API (Amid et al., 2020). As a result of the International Nucleotide Sequence Database Collaboration (INSDC), ENA and NCBI both host BioProject accessions (Arita et al., 2021).

To vizualise and cluster the data, we computed the nearest neighbors distance matrix and a neighborhood graph of observations and applied Scanpy's reproducible, single threaded UMAP dimensionality reduction implementation (McInnes et al., 2018). We clustered the neighborhood graph using the Leiden algorithm (Traag et al., 2019). To select an optimal Leiden clustering resolution, we sequentially applied Leiden clustering to the neighborhood graph at a range of resolutions from 0.1–2.0 at intervals of 0.1, computed the Jaccard index (also known as "intersection over union" or "IoU") of each CZ CELLxGENE `cell_type`-Leiden cluster pair, and determined the optimal one-to-one matching using a modified Jonker–Volgenant algorithm as implemented by SciPy's `linear_sum_assignment` function (Crouse, 2016; Virtanen et al., 2020). Because the function minimizes total assignment cost, we supplied the negated Jaccard matrix and summed the original Jaccard indices at the returned matches. We retained the Leiden clustering resolution with the highest matched Jaccard sum as the canonical Leiden clustering resolution.

To test our clustering methodology, we implemented it as a set of importable functions and applied them to datasets corresponding with single SRA experiment accessions before applying it to the atlas. We drew one stratified sample of 100,000 cells for calibration and a separate stratified sample of 100,000 cells for validation. The validation sample was drawn using a different random seed from the calibration sample. For both samples, the strata were BioProjects, and the number of cells drawn from each stratum was proportional to that BioProject's share of the full atlas, with at least one cell per BioProject. We benchmarked the batch key using scIB (Luecken et al., 2022).

Results and discussion

There were 35,266 datasets available in the `2026-01-12` release of scBaseCount. We discarded 472 accessions whose identifiers were not SRA or ENA records. We retained accessions whose free-text tissue label matched a lung-related regular expression, leaving 2,396 lung-tissue accessions. After retrieving study accessions from the ENA portal API, we dropped 573 of those accessions from seven multi-organ studies, and any accession that lacked a study accession. We then dropped studies whose combined cell count was 1,000 or fewer, removing 7 further accessions. The remaining 1,816 accessions (13,013,643 cells; mean 7,166, median 5,112 cells per accession) formed the eligible set.

The atlas contained 1,410 datasets from 1,816 eligible datasets, spanning 172 BioProject accessions. All 1,816 eligible datasets reached per-cell QC (13,013,643 barcodes). After those cell filters, 10,893,794 barcodes remained. File-level gates then excluded 406 datasets, and concatenation retained 9,307,963 barcodes (Table 1). Restricted to the 1,410 concatenated datasets, the same cell filters removed 578,434 of 9,886,397 barcodes (145,320 below 200 genes, 430,729 at or above 20% mitochondrial counts, and 2,385 at or above 50% ribosomal counts). The atlas contained 61 CZ CELLxGENE `cell_type` labels (Fig 1).

Table 1. Atlas contained 1,410 datasets (9,307,963 barcodes) across 172 BioProject accessions, of 1,816 eligible datasets (13,013,643 barcodes) following filtering.

Filter	Datasets dropped, n (%)	Barcodes dropped, n (%)
MD5 hash (mismatch)	0 (0)	0 (0)
CZ CELLxGENE `cell_type` (labels absent)	238 (13.1)	1,509,128 (11.6)
Min. genes per cell (< 200)	—	1,600,056 (12.3)
Max. mitochondrial fraction ($\geq$ 20%)	—	517,018 (4.0)
Max. ribosomal fraction ($\geq$ 50%)	—	2,775 (0.02)
Min. cells remaining (< 100)	100 (5.5)	1,169 (0.01)
Min. fraction of cells remaining (< 50%)	68 (3.7)	75,534 (0.58)

Dataset percentages use the 1,816 eligible datasets as the denominator. Barcode percentages use the 13,013,643 barcodes present in those datasets before QC. Per-cell filter counts include every file that reached QC, including files later rejected by file-level gates; they are not restricted to concatenated files. A total of 406 datasets were dropped. Of those, 0 were dropped due to MD5 hash mismatch, 238 (13.1% total datasets) were dropped due to fully absent CZ CELLxGENE `cell_type` labels. Of the 3,705,680 barcodes that were dropped from the eligible set, 1,600,056 (12.3% total barcodes) were dropped for gene counts of less than 200, 517,018 (4.0% total barcodes) were dropped for having 20% or more of total gene count being mitochondrial, and 2,775 (0.02% total barcodes) for having 50% or more of total gene counts being ribosomal. Following cell filtering, 100 (5.5% total datasets) were dropped for retaining fewer than 100 cells, and 68 (3.7% total datasets) were dropped for retaining less than 50% of cells. Among concatenated files only, cell filters dropped 578,434 barcodes.

On atlas scale, batch effect removal per CZ CELLxGENE `cell_type` label, measured by the k-nearest-neighbor batch effect test (kBET) proposed by Büttner et al. (2019) and implemented by Luecken et al. (2022) was lower from the uncorrected PCs compared to the batch corrected PCs, the latter using

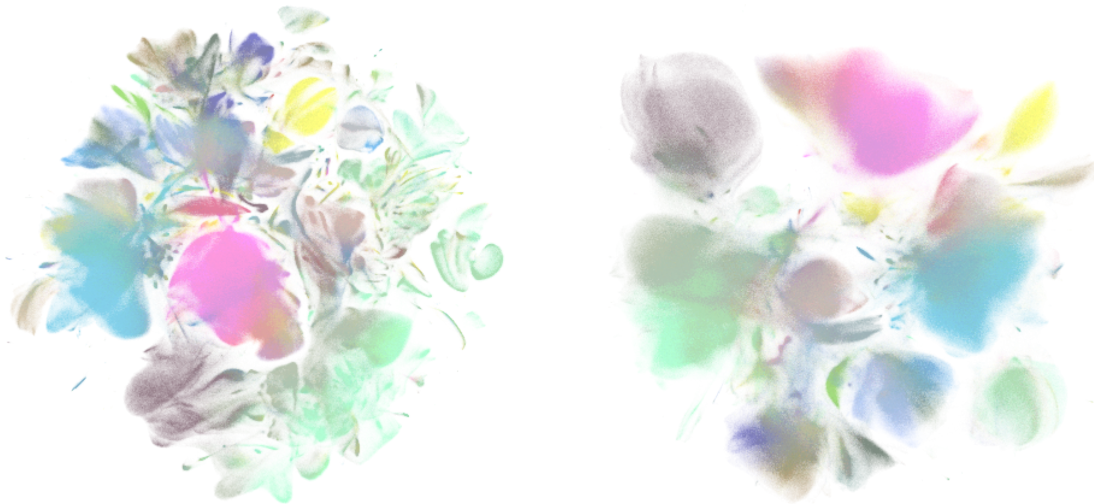

Fig 1. Single-file integrated lung disease atlas derived from scBaseCount. UMAP of the integrated atlas colored by CZ CELLxGENE `cell_type`. A: Uncorrected embedding. B: Harmony-corrected embedding (batch key: BioProject).

NCBI BioProject as batch key (TODO:decide whether to cite all scib metrics in text). The aggregate biological conservation score proposed by Luecken et al. (2022) decreased from 0.53 to 0.51 as a result of batch correction, while the batch correction aggregate score increased from 0.28 to 0.45 (Fig 2).

Those two aggregates moved in opposite directions and by different amounts: batch correction rose by 0.17, while biological conservation fell by 0.02 (Fig 2). Harmony takes the PC coordinates of cells and attempts to correct them by moving cells to favor mixed representation of cells in clusters, while maintaining biological groupings (Korsunsky et al., 2019). Therefore, it is not surprising that we observed a gain of large magnitude on batch correction metrics, since the evaluation key is the same BioProject column used for correction. That gain is therefore not independent evidence that BioProject is the uniquely correct batch definition. However, cell-type conservation, which is scored against CZ CELLxGENE `cell_type`, exhibited minimal change. Other atlas collections can of course yield a different balance. Independent benchmarks place Harmony among methods that perform similarly to single-cell foundation models such as scGPT and Geneformer on batch correction, and indeed more stable than foundation models when they are used zero-shot, or without fine-tuning for batch correction (Cui et al., 2024; Kedzierska et al., 2025; Liu et al., 2026; Theodoris et al., 2023).

We validated our resolution selection methodology on five lung experiment files from the same release, sampled at the 25th, 33rd, 50th, 67th, and 75th percentiles of cell count (3,413–9,419 cells before filtering; Fig 3 highlights the 67th-percentile file). On each file we ran Leiden clustering at resolutions 0.1–2.0 in steps

Method	Bio conservation					Batch correction					Aggregate score		
	Isolated labels	KMeans NMI	KMeans ARI	Silhouette label	cLISI	BRAS	iLISI	KBET	Graph connectivity	PCR comparison	Batch correction	Bio conservation	Total
X_pca_harmony	0.45	0.46	0.20	0.44	0.99	0.75	0.02	0.48	0.47	0.52	0.45	0.51	0.49
X_pca	0.50	0.51	0.21	0.45	0.99	0.50	0.00	0.31	0.60	0.00	0.28	0.53	0.43

Fig 2. Batch correction metrics before and after Harmonization of PCs. Benchmark results on a stratified atlas subset comparing uncorrected PCs and Harmony-corrected PCs (batch key: NCBI BioProject). Aggregate biological conservation decreased from 0.53 to 0.51; aggregate batch correction increased from 0.28 to 0.45. kBET was lower for uncorrected than batch-corrected embeddings.

of 0.1, scored each partition by the summed Jaccard index of the optimal one-to-one matching between clusters and CZ CELLxGENE `cell_type` labels, and retained the resolution with the highest score. Selected resolutions ranged from 0.1 to 1.5, and cluster counts were close to the number of retained cell-type labels. Silhouette scores generally fell as resolution increased, and the Jaccard-selected resolution was never the silhouette maximum: silhouette rewards compact, well-separated groups, whereas the cell-type prior often sits inside larger partitions. On a 100,000-cell, study-stratified subset of the atlas the same matching criterion favoured resolution 0.8 (21 clusters in the subset). We used that resolution for the Harmony-corrected atlas, which produced 71 clusters.

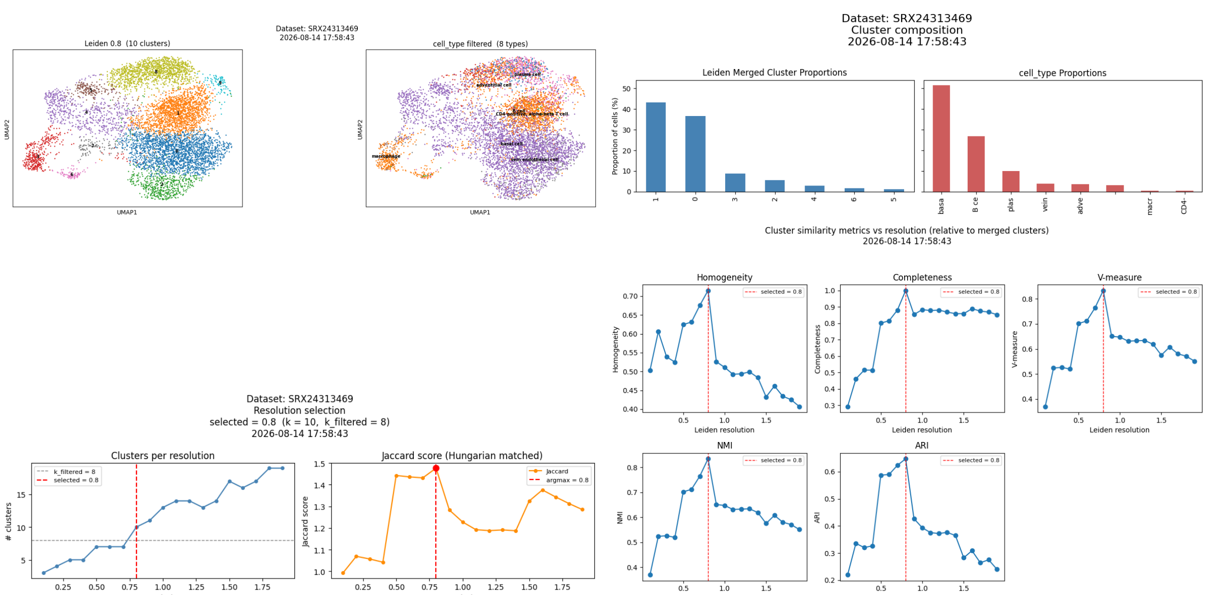

Fig 3. Representative Leiden resolution sweep from the five-file validation (SRX24313469, 67th percentile of cell count). A: UMAP of Leiden clusters at the Jaccard-selected resolution versus CZ CELLxGENE `cell_type` labels. B: Cluster and cell-type composition. C: Cluster count and matched Jaccard score across resolutions 0.1–2.0. D: Homogeneity, completeness, V-measure, NMI, and ARI versus resolution. The selected resolution was 0.8.

Continued study is warranted using the present atlas. Due to the atlas spanning over 100 BioProject accessions, we recognize that differential gene expression analyses on various clusters has the potential to reaffirm disease state gene expression in certain cell types, or to highlight novel, low signal relationships that have the potential to represent new findings. As evident in the UMAP projection of the atlas, certain CZ CELLxGENE `cell_type` labels tend to cluster naturally, such as B and T lymphocytes (Fig 1 B). Sant et al. (2025) note that a cluster's ability to reliably representat real cellular identity and state is crucial for deriving biological meaning in downstream analysis steps. Since the clustering of the present atlas builds on the CZ CELLxGENE `cell_type` annotations as priors informing the clustering resolution, the clusters may indeed represent real biology.

Furthermore, we identified limitations with the study. Starting with atlas construction, reproduction requires cloud storage in the order of hundreds of GB. The concatenation of `h5ad` files was optimized for limited disk space rather than limited memory, and therefore requires that the entire concatenated atlas is held in memory before it is written to disk. Realistically, memory in the order of TB is required to safely reproduce the atlas or build a new one with the code present to avoid out-of-memory errors late in the concatenation. Additionally, given that scBaseCount is agentically curated, it is not surprising that we identified certain `h5ad` files that upon close inspection using the ENA portal API did not appear to entirely match the metadata perscribed by the agent built by Youngblut et al. (2025). While we sought out to identify reliably accurate fields in our deterministic context lookup on ENA via SRA experiment accession, it is entirely possible that certain metadata fields we rely on for context lookup can be missing or inaccurate, which can negatively affect downstream analysis, which relies on assumptions about which cells are in which state. Furthermore, there is potential to deepen the methodology of data processing. This is particularly true for the selection of HVGs. Multiple methods exist for ranking genes, and the number of genes plays an important role in the downstream granularity of the cell embedding, as well as with the materialization of batch effects (Kiselev et al., 2019; Li et al., 2026; Zhao et al., 2025). Neither HVG selection methodology nor HVG count were optimized in the methodology presented.

Conclusion

The field of single-cell technology continues to grow rapidly. With it, the availability of data, tools, and analysis methodology, has become difficult to navigate (Heumos et al., 2023). Meanwhile, expanded cell counts in scRNA-seq datasets is crucial for executing analysis of complex cell populations with rare cell types and states (Kolodziejczyk et al., 2015). Nonetheless, simply increasing the number of cells considered in scRNA-seq data analysis is not sufficient for generating those large datasets (Mueller et al., 2026). To produce processed datasets at large scale represents a challenge spanning data aquisition, processing, and correction for batch effects that materialize at atlas-level. Here, we propose a method for identifying and utilizing publically available resources in the form of large raw matrix stores—in this case scBaseCount—and integrating data to produce a single, analysis-ready atlas with extensive processing applied (Youngblut et al., 2025). The atlas supplies usable clusters, which the source files lacked.

The atlas building pipeline proposed here does not exist in isolation. During the writing of the present report, a Snakemake workflow for concatenating raw count matrices, and applying quality control and batch correction was developed by Mueller et al. (2026), a method that addresses the growing demand for curated atlases to be used for downlstream data analysis. Continued work would certainly be warranted in implementing the pipeline built by Mueller et al., 2026, as a means of validating the quality control and integration decisions made in the present method or producing more atlases.

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